Supplementary Material for: Scalable Throughput Estimation for Large-Scale Optical Networks via Column Generation

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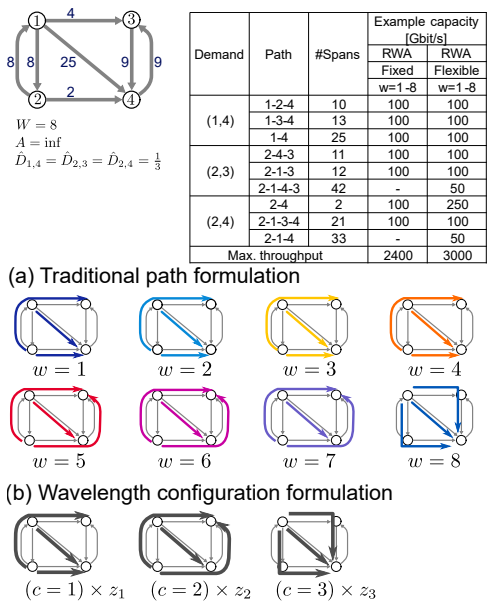
I. Small Example

We consider the RWA problem in a 4-node network shown in Fig. 1. There are 8 wavelengths. Each link has two fibers that support bidirectional traffic. Three demands $(1,4)$, $(2,3)$, and $(2,4)$ are assumed, and the traffic demand is uniformly distributed. We discuss the number of variables required to optimize the lightpaths for these demands via two approaches, the Integer Linear Programming (ILP) model and the Column Generation (CG) approach. Note that both approaches have assumed K different paths for each node pair, $K = 3$. Assuming that the lightpaths have distance-adaptive transmission capacity:

- $(1,4) : \{p_{124}, p_{134}, p_{14}\}$ with distance-adaptive transmission capacity $C_{p_{124}} = 100, C_{p_{134}} = 100, C_{p_{14}} = 100$.
- $(2,3) : \{p_{243}, p_{213}, p_{2143}\}$ with distance-adaptive transmission capacity $C_{p_{243}} = 100, C_{p_{213}} = 100, C_{p_{2143}} = 50$.
- $(2,4) : \{p_{24}, p_{2134}, p_{214}\}$ with distance-adaptive transmission capacity $C_{p_{24}} = 250, C_{p_{2134}} = 100, C_{p_{214}} = 50$.



Demand	Path	#Spans	Example capacity [Gbit/s] RWA Fixed w=1-8	Example capacity [Gbit/s] RWA Flexible w=1-8
(1,4)	1-2-4	10	100	100
	1-3-4	13	100	100
	1-4	25	100	100
(2,3)	2-4-3	11	100	100
	2-1-3	12	100	100
	2-1-4-3	42	-	50
(2,4)	2-4	2	100	250
	2-1-3-4	21	100	100
	2-1-4	33	-	50
Max. throughput			2400	3000

Fig. 1. Illustration of a 4-node network with link number showing the length in multiples of 100 km, $W = 8$.

A. Integer Linear Programming Model

We use $p_{s,d,k}$ to denote the kth route between two nodes s and d. Let $\delta_{s,d,k,w}$ be the indicator of whether the lightpath $p_{s,d,k}$ is assigned to wavelength w, and $T_{s,d}$ represent the total transmission capacity for the lightpaths connecting (s,d). Thus, we have $T_{s,d} = \sum_k \sum_w \delta_{s,d,k,w} C_{s,d,k}$.

a) Parameters:

- $p_{s,d,k}$: path connecting (s,d) on the k-th route. It may also be simply noted as $p_{s..d}$ with the route subscript.
- $C_{s,d,k}$: transmission capacity of path $p_{s,d,k}$.

b) Variables:

- $\delta_{s,d,k,w}$: equals 1 if path $p_{s,d,k}$ uses a wavelength w, 0 otherwise.
- T_N : network throughput.

$$
\begin{aligned}
\max_{\delta_{s,d,k,w}\in\{0,1\}} \quad & T_N \\
\textbf{s.t.} \quad & \frac{1}{3}T_N + (-100)\delta_{1,4,p_{124},1} + (-100)\delta_{1,4,p_{134},1} + (-100)\delta_{1,4,p_{14},1} + (-100)\delta_{1,4,p_{124},2} + \cdots \leq 0 && \text{(1a)}\\
& \frac{1}{3}T_N + (-100)\delta_{2,3,p_{243},1} + (-100)\delta_{2,3,p_{213},1} + (-50)\delta_{2,3,p_{2143},1} + (-100)\delta_{2,3,p_{243},2} + \cdots \leq 0 && \text{(1b)}\\
& \frac{1}{3}T_N + (-250)\delta_{2,4,p_{24},1} + (-100)\delta_{2,4,p_{2134},1} + (-50)\delta_{2,4,p_{214},1} + (-250)\delta_{2,4,p_{24},2} + \cdots \leq 0 && \text{(1c)}\\
& \delta_{1,4,p_{124},1} \leq 1, \qquad (\ell = \ell_{12}, w = 1) && \text{(1d)}\\
& \delta_{1,4,p_{134},1} + \delta_{2,3,p_{213},1} + \delta_{2,4,p_{2134},1} \leq 1, \qquad (\ell = \ell_{13}, w = 1) && \text{(1e)}\\
& \delta_{1,4,p_{14},1} + \delta_{2,3,p_{2143},1} + \delta_{2,4,p_{214},1} \leq 1, \qquad (\ell = \ell_{14}, w = 1) && \text{(1f)}\\
& \cdots \\
& \delta_{2,4,p_{214},8} \leq 1, \qquad (\ell = \ell_{21}, w = 8) && \text{(1g)}\\
& \delta_{2,3,p_{243},8} + \delta_{2,3,p_{2143},8} \leq 1, \qquad (\ell = \ell_{43}, w = 8) && \text{(1h)}
\end{aligned}
$$

The objective is to maximize the network throughput, subject to the uniformly distributed network demand profile. Note that the constraints of this model are written following a standard linear programming form of $\max\{cx \mid Ax \leq b, x \geq 0\}$. Constraints (1a)-(1c) ensure that the traffic demand distribution of $(1,4)$, $(2,3)$, and $(2,4)$ is compliant with the network demand profile. Constraints (1d)-(1h) ensure no resource conflict for each wavelength and for each link. The maximal T_N of 3000 Gbps is attained by optimizing 72 boolean variables, $\delta_{s,d,k,w}$. One possible assignment is shown in Fig. 1(a), where we see three paths p_{14}, p_{213}, and p_{24} on the 1st wavelength. This means $\delta_{1,4,3,1}(\delta_{1,4,p_{14},1})$, $\delta_{2,3,2,1}(\delta_{2,3,p_{213},1})$, and $\delta_{2,4,1,1}(\delta_{2,4,p_{24},1})$ are 1, whereas the other indicators on the 1st wavelength are all 0.

B. Numerical Example of the Column Generation Algorithm

Let us now return to the 4-node example to show the numerical computations with CG. The configuration set is initialized with the following three simple configurations.

- $c_1 = ([p_{124}], [], []) \rightarrow \vec{T_1} = [100; 0; 0]$,
- $c_2 = ([], [p_{243}], []) \rightarrow \vec{T_2} = [0; 100; 0]$,
- $c_3 = ([], [], [p_{24}]) \rightarrow \vec{T_3} = [0; 0; 250]$

Iteration 1

$$
\begin{aligned}
\max_{z_c\in\mathbb{R}^+} \quad & T_N \\
\textbf{s.t.} \quad & \frac{1}{3}T_N - 100z_1 - 0z_2 - 0z_3 \leq 0 && [\sigma_{1,4}] && \text{(2a)}\\
& \frac{1}{3}T_N - 0z_1 - 100z_2 - 0z_3 \leq 0 && [\sigma_{2,3}] && \text{(2b)}\\
& \frac{1}{3}T_N - 0z_1 - 0z_2 - 250z_3 \leq 0 && [\sigma_{2,4}] && \text{(2c)}\\
& z_1 + z_2 + z_3 \leq 8 && [\sigma_W] && \text{(2d)}
\end{aligned}
$$

This RMP is relaxed, so that we can obtain the relaxed upper bound of $T_N^* = 1000$ by the optimal solutions $\vec{z}^*$=$[\frac{10}{3}, \frac{10}{3}, \frac{4}{3}]$. The parameters $\sigma_{s,d}$ and σ_W can be easily retrieved by any optimization solver. In this example, the optimal dual solution is $\sigma = [\sigma_{1,4}, \sigma_{2,3}, \sigma_{2,4}] = [\frac{5}{4}, \frac{5}{4}, \frac{1}{2}]$, $\sigma_W = 125$.

We then solve the pricing problem.

$$\xi^1 = \max_{\delta_{s,d,p}\in\{0,1\}} (T_{1,4,4})\frac{5}{4} + (T_{2,3,4})\frac{5}{4} + (T_{2,4,4})\frac{1}{2} - \sigma_W$$

$$\begin{aligned}
\textbf{\textit{s.t.}}\quad & T_{1,4,4} = 100\delta_{1,4,p_{124}} + 100\delta_{1,4,p_{134}} + 100\delta_{1,4,p_{14}} && \text{(3a)}\\
& T_{2,3,4} = 100\delta_{2,3,p_{243}} + 100\delta_{2,3,p_{213}} + 50\delta_{2,3,p_{2143}} && \text{(3b)}\\
& T_{2,4,4} = 250\delta_{2,4,p_{24}} + 100\delta_{2,4,p_{2134}} + 50\delta_{2,4,p_{214}} && \text{(3c)}\\
& \delta_{1,4,p_{124}} \le 1 \qquad (\ell = \ell_{12}) && \text{(3d)}\\
& \delta_{1,4,p_{134}} + \delta_{2,3,p_{213}} + \delta_{2,4,p_{2134}} \le 1 \qquad (\ell = \ell_{13}) && \text{(3e)}\\
& \delta_{1,4,p_{14}} + \delta_{2,3,p_{2143}} + \delta_{2,4,p_{214}} \le 1 \qquad (\ell = \ell_{14}) && \text{(3f)}\\
& \delta_{1,4,p_{124}} + \delta_{2,3,p_{243}} + \delta_{2,4,p_{24}} \le 1 \qquad (\ell = \ell_{24}) && \text{(3g)}\\
& \delta_{1,4,p_{134}} + \delta_{2,4,p_{2134}} \le 1 \qquad (\ell = \ell_{34}) && \text{(3h)}\\
& \delta_{2,3,p_{213}} + \delta_{2,3,p_{2143}} + \delta_{2,4,p_{2134}} + \delta_{2,4,p_{214}} \le 1 \qquad (\ell = \ell_{21}) && \text{(3i)}\\
& \delta_{2,3,p_{243}} + \delta_{2,3,p_{2143}} \le 1 \qquad (\ell = \ell_{43}) && \text{(3j)}
\end{aligned}$$

We find the first promising configuration with the reduced cost $\xi^1 = 250$ and $\delta^* = [\delta_{p_{124}}, \delta_{p_{134}}, \delta_{p_{14}}, \delta_{p_{243}}, \delta_{p_{213}}, \delta_{p_{2143}}, \delta_{p_{24}}, \delta_{p_{2134}}, \delta_{p_{214}}] = [1, 1, 1, 0, 0, 0, 0, 0, 0]$. As $\xi^1 \ge 0$, continue and add the configuration $c_4 = ([p_{124}, p_{134}, p_{14}], [], [])$ into the RMP with the transmission capacities $\vec{T}_4 = [300; 0; 0]$.

Iteration 2

RMP. The primal solution $T_N^* = \frac{18000}{13}$, $\vec{z}^* = [0, \frac{60}{13}, \frac{24}{13}, \frac{20}{13}]$, and the dual solution $\sigma = [\frac{15}{26}, \frac{45}{26}, \frac{9}{13}]$, $\sigma_W = \frac{2250}{13}$.
Pricing problem. The objective function is now $\xi^2 = \max \frac{15}{26}(+T_{1,4,5}) + \frac{45}{26}(+T_{2,3,5}) + \frac{9}{13}(+T_{2,4,5}) - \frac{2250}{13}$. Optimal solution $\xi^2 = \frac{3000}{13}$, $\delta^* = [0, 0, 1, 0, 1, 0, 1, 0, 0]$. As $\xi^2 \ge 0$, continue and add the configuration $c_5 = ([p_{14}], [p_{213}], [p_{24}])$ into the RMP with $\vec{T}_5 = [100; 100; 250]$.

Iteration 3

RMP. The primal solution $T_N^* = 2400$, $\vec{z}^* = [0, 0, 0, 0, 8]$, and the dual solution $\sigma = [0, 3, 0]$, $\sigma_W = 300$.
Pricing problem. Optimal solution $\xi^3 = 300$, $\delta^* = [0, 0, 0, 1, 1, 0, 0, 0, 0]$. As $\xi^3 \ge 0$, continue and add the configuration $c_6 = ([], [p_{243}, p_{213}], [])$ into the RMP with $\vec{T}_6 = [0; 200; 0]$.

Iteration 4

RMP. The primal solution $T_N^* = \frac{8000}{3}$, $\vec{z}^* = [0, 0, 0, \frac{16}{9}, \frac{32}{9}, \frac{8}{3}]$, and the dual solution $\sigma = [\frac{10}{9}, \frac{5}{3}, \frac{2}{9}]$, $\sigma_W = \frac{1000}{3}$.
Pricing problem. Optimal solution $\xi^4 = \frac{1000}{9}$, $\delta^* = [0, 0, 1, 1, 1, 0, 0, 0, 0]$. As $\xi^4 \ge 0$, continue and add the configuration $c_7 = ([p_{14}], [p_{243}, p_{213}], [])$ into the RMP with $\vec{T}_7 = [100; 200; 0]$.

Iteration 5

RMP. The primal solution $T_N^* = 3000$, $\vec{z}^* = [0, 0, 0, 1, 4, 0, 3]$, and the dual solution $\sigma = [\frac{5}{4}, \frac{5}{4}, \frac{1}{2}]$, $\sigma_W = 375$.
Pricing problem. Optimal solution $\xi^5 = 0$, $\delta^* = [1, 1, 1, 0, 0, 0, 0, 0, 0]$. As $\xi^5 = 0$, stop.

As the solution $\vec{z}^* = [0, 0, 0, 1, 4, 0, 3]$ is already an integer, the original problem is solved. If $\vec{z}^*$ is not an integer, we shall pass the existing columns back to RMP and resort to a branch-and-bound algorithm to obtain the integer solution [1].

The ILP model of the main text has a block-matrix structure, which confirms its decoupling into isolated wavelength configurations. This structure is illustrated in Fig. 2, where we plot the nonzero entries in the coefficient matrix. In Fig. 2(a), eight identical diagonal blocks are colored in blue. Each of these blocks has the same nonzero pattern. The presence of these blocks indicates that the routing for these wavelengths shares an identical solution space. This allows us to repeatedly solve the same pricing problem to price out the new configurations. These configurations are aggregated in the master problem solely through the wavelength limitation constraint, which is visualized in the fourth row of Fig. 2(b).

References

[1] L. A. Wolsey, *Integer Programming*, 2nd ed. Hoboken, NJ, USA: Wiley, 2020.

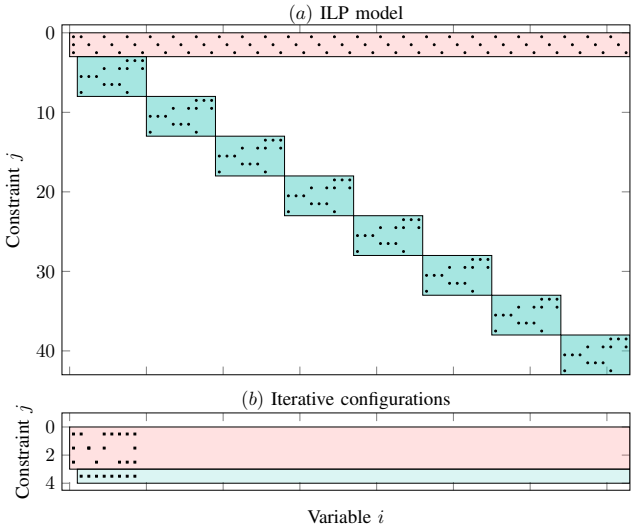


Fig. 2. Nonzero entries in the ILP model's coefficient matrix. The above model still considers the 4-node network with 8 wavelengths and 3 demands. (a) is formulated through traditional path-formulation, while (b) and (c) are formulated through the wavelength configuration formulation. (a) has $73(= 3\times3\times8+1)$ variables and $43(= 5 \times 8 + 3)$ constraints, (b) has 4 variables and 4 constraints, and (c) has 9 variables and 4 constraints. The red blocks indicate the demand-capacity constraints, and blue blocks show the wavelength non-overlapping constraints reformulated as a linking constraint.